

MSME IDEA HACKATHON 6.0



Objective: MSME Idea Hackathon 6.0 aims to foster innovation in frontier technologies by enabling MSMEs, startups, and innovators to develop scalable, cost-effective, and sustainable solutions across domains such as renewable energy, energy efficiency, automotive advancements, Industry 4.0/5.0, robotics, and emerging export-oriented sectors. The initiative seeks to strengthen global competitiveness, and accelerate the creation of future-ready, self-reliant MSME ecosystems aligned with Aatmanirbhar Bharat and Viksit Bharat @2047.

Target Group: The Hackathon is open to all India citizens aged above 18 years of age, regardless of their professional background or educational qualification. Participants may include MSMEs, students, entrepreneurs, technocrats, working professionals, academic researchers, and retired experts. The only prerequisite is a passion for problem-solving and a commitment to nation building. Participants shall apply on MSME Innovative portal on www.innovative.msme.gov.in. Last date to apply for Hackathon 6.0 is 14.07.2026

Frontier Technology in MSMEs: One Pillar, Six Themes

Themes for MSME Idea Hackathon 6.0

1. Renewable Energy- To promote the generation, storage, management and productive use of renewable energy through MSME-led products and technologies. Participants may work on solar, wind, biomass, biogas, bioenergy, green hydrogen, hybrid energy systems, battery storage, second-life batteries, renewable-energy-powered machinery, decentralized energy systems or clean-energy components. Solutions may include solar dryers, cold-storage units, pumps, process-heating systems, compact storage devices, monitoring tools or indigenous components that improve energy access and create commercially viable clean-technology opportunities for MSMEs.

2. Energy Efficiency- This sub-theme invites practical and affordable ideas that help MSME units reduce energy consumption, production costs and process losses. Participants may propose machine-level energy-monitoring devices, efficient motors and drives, compressed-air leakage detection, waste-heat recovery, efficient boilers and furnaces, improved refrigeration and cooling systems, process-

optimization tools, thermal insulation, power-quality improvement or low-cost energy-solutions. Ideas should preferably demonstrate measurable energy savings and should be suitable for retrofitting or adoption by existing MSME units with limited capital and technical capacity.

3. Automotive Technology- This sub-theme seeks innovative ideas and prototype-ready solutions addressing emerging requirements across automotive manufacturing, electric mobility and mobility-support ecosystems. Participants may develop indigenous automotive components, electric motors and controllers, battery-management and thermal-management systems, charging and battery-swapping technologies, vehicle electronics, embedded systems, sensors, connected-vehicle solutions, lightweight and advanced materials, safety technologies, alternative-fuel systems, diagnostic equipment, testing tools and precision-manufactured parts. Ideas may also address component remanufacturing, vehicle maintenance, end-of-life vehicle recovery and digital solutions for automotive suppliers.

4. Industry 4.0 and 5.0- This sub-theme invites ideas that can make MSME manufacturing more connected, intelligent, flexible and human-centric through technologies such as the Internet of Things, digital twins, artificial intelligence, machine vision, cyber-physical systems and smart production monitoring. Participants may develop affordable sensors, machinery-retrofitting solutions, predictive-maintenance tools, digital work instructions, production, worker-assistance systems, product traceability solutions or other technologies that allow MSMEs to modernize existing operations without requiring complete replacement of their machinery and infrastructure.

5. Robotics and Automation- This sub-theme seeks affordable and adaptable automation solutions designed specifically for the scale and operating conditions of MSMEs. Participants may develop pick-and-place systems, machine-tending solutions, automated inspection tools, robotic welding or finishing systems, material-handling equipment, autonomous factory vehicles, computer-vision-based automation, modular controllers, adaptive grippers or retrofit kits for conventional machinery. Solutions should focus on ease of installation, safety, affordability, simple programming and suitability for small-batch, high-variety or labour-intensive production environments.

6. Other Frontier Technologies- This sub-theme invites innovative ideas in emerging and high-growth technology sectors not fully covered under the preceding categories, including advanced materials, drones, space technology, defence and dual-use technologies, medical devices, electronics and other high growth sectors. Participants may propose indigenous components, advanced coatings and composites, drone payloads and control systems, satellite-enabled applications, portable defence or communication systems, affordable diagnostic and assistive devices, embedded electronics, precision-engineered products, testing technologies or other solutions with clear potential for domestic manufacturing, import substitution, commercialization and exports.